

An Exact Computer-Assisted Improvement of the Lower Bound for Packing Eleven Unit Squares

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Abstract

We give a computer-assisted certificate implying the strict lower bound

$$s(11) > \frac{3,875,000,000}{999,999,999} = 3.875000003875000003875\dots$$

for packing eleven independently rotatable unit squares into a square. The construction starts from Kleddamag's September 2026 parent-core threshold certificate for $s(11) > 31/8$. We first reweight thirteen threshold-charge orbits, increasing the exact counting surplus from 107,864 to 428,125 units of 10^{-9} while keeping the same geometric parameters. We then multiply the parent side A and every row-specific strict-core side B_r by the common rational factor $\lambda = 999,999,999/1,000,000,000$. The strict parent-to-core containment inequalities are preserved by this common scaling, while the enlarged parent-centre domains and smaller cores make coverage weakly harder and are therefore rechecked. Two complete exact implementations replay all 12,028 parent-angle intervals and both return the same minimum charge, 1,000,047,559 units, on every interval. Since eleven times this minimum exceeds the total feature budget 11,000,095,024 by 428,125 units, the parent-core counting theorem excludes eleven parents of side A' in a container of side $L = 191/50$, yielding the stated strict ratio L/A' . A public registry check on 29 September 2026 found that the Squares Project still listed $31/8$ as the strongest verified lower bound for $n = 11$; no stronger registered lower bound was located in that check.

1 Statement of the result

Let $s(n)$ denote the least side length of a square that contains n pairwise interior-disjoint unit squares, each allowed an independent rotation. We certify:

Theorem.

$$s(11) > \frac{3,875,000,000}{999,999,999} = 3.875000003875000003875\dots$$

The improvement over $31/8$ is exact:

$$\frac{3,875,000,000}{999,999,999} - \frac{31}{8} = \frac{31}{7,999,999,992} > 0.$$

The result is deliberately modest numerically: the contribution is not a new packing and does not determine $s(11)$. Its significance is that the existing 31/8 certificate contains enough hidden slack, after reweighting, to support a strictly stronger exact parent scaling.

2 Relation to the previous 31/8 certificate

The starting point is Kleddamag’s public September 2026 certificate, itself developed from the Squares Project threshold-certificate framework. Its frozen source data use

$$L = \frac{191}{50}, \quad A = \frac{764}{775}, \quad \frac{L}{A} = \frac{31}{8}.$$

The source certificate contains 12,028 parent-angle intervals. It assigns to every legal parent square a concentric strict interior core and defines a nonnegative logical charge from weighted sites and threshold features. Across pairwise disjoint cores, each feature has a finite combinatorial budget. If Γ is a universal lower bound on core charge and M is the total budget, then $11\Gamma > M$ rules out eleven side- A parents in side L ; scaling back gives a strict lower bound L/A for unit squares.

Quantity	Source 31/8	Reweighted	Change
Minimum charge Γ (10^{-9} units)	999,962,528	1,000,047,559	+85,031
Budget M (10^{-9} units)	10,999,479,944	11,000,095,024	+615,080
Counting surplus $11\Gamma - M$	107,864	428,125	+320,261
Angle intervals	12,028	12,028	unchanged

3 Fixed-parameter reweighting

Only thirteen charge-orbit weights are changed. All changes are nonnegative integer increments in the certificate’s 10^{-9} weight units, so feature nonnegativity is preserved.

Orbit	Increment	New weight
3	+1	863,819
27	+4	370,824
43	+3	9,293,475
51	+4,384	1,928,668
52	+7	5,679,796
130	+34,344	14,245,863
131	+1	4,083,536
174	+3,776	881,863
206	+5	19,257,667
207	+1	58,656
218	+7	9,486,761
232	+5	15,656
259	+1	153,072

A complete exact scan at the original geometric parameters gives $\Gamma = 1,000,047,559$ on every one of the 12,028 rows. This converts the source certificate’s highly uneven low rows into a flat minimum plateau and increases the counting surplus by a factor of about four. This step alone does not improve the lower-bound ratio because L/A is still 31/8.

4 Rational parent/core scaling

We next fix the weights and apply the common factor

$$\lambda = \frac{999,999,999}{1,000,000,000} = 1 - 10^{-9}$$

to the parent side A and to every row-specific core side B_r . Thus

$$A' = \lambda A = \frac{190,999,999,809}{193,750,000,000}, \quad B'_r = \lambda B_r.$$

The source strict-containment test has the form $A > BW$, where W is the maximum relative width over the row’s orientation interval. After the common scaling it becomes $\lambda A > \lambda BW$. Because $\lambda > 0$, every strict parent-to-core containment inequality remains strict. The difficult part is coverage: smaller cores and the corresponding legal parent-centre geometry can create new low-charge cells. For that reason no continuity or floating-point argument is used; all 12,028 rows are replayed exactly.

The resulting ratio is

$$\frac{L}{A'} = \frac{3,875,000,000}{999,999,999}.$$

5 Exact verification

5.1 Primary complete replay

The primary exact implementation uses rational geometry and integer event-cell sweeping. It checks every legal centre cell in each of the 12,028 parent-angle rows. At the scaled parameters A' and B'_r it reports

$$\min_r \Gamma_r = \max_r \Gamma_r = 1,000,047,559, \quad r = 0, \dots, 12,027.$$

Hence the complete histogram consists of a single value with multiplicity 12,028. The exact budget is unchanged by geometric scaling:

$$M = 11,000,095,024, \quad 11\Gamma - M = 428,125 > 0.$$

5.2 Independent full replay

A second implementation, written independently of the primary verifier, reconstructs the same certificate semantics using Python `Fraction`/integer geometry, exact integer cross-product legality tests, and a separately implemented block range-minimum sweep. Its complete replay of rows 0–12,027 also reports the one-bin histogram

$$\{1,000,047,559 : 12,028\},$$

with zero bad rows. The recorded run took approximately 659.6 seconds. This implementation shares the same frozen certificate data and mathematical definitions, but does not import the primary verifier.

5.3 Frozen hashes and receipts

Source certificate SHA-256:

```
57e9927da5c13f42dd8bcbf8f08c84363635fece626657ee63a810c61cd44458
```

Improved derived certificate SHA-256:

```
31e10ceb8368cc858e61f45ce7cfee783e8d0c7910893559164810f45439cc34
```

The release packet includes the frozen improved certificate, the fixed-parameter reweighting receipt, both recorded full-replay receipts, a portable copy of Kleddamag’s MIT-licensed primary Python sweep, and the separately implemented independent verifier used for the second complete replay.

6 Proof of the stated bound

The mathematical implication from the finite certificate is the same parent-core counting argument used by the source certificate. We include it here to make the new ratio explicit.

Charge budget. For every family of pairwise disjoint closed cores, the sum of the logical charges is at most the feature budget M . Ordinary point features have budget w , while a k -of- m threshold feature of weight w has budget $\lfloor m/k \rfloor w$. Budgets add even when different features share sites.

Strict transport. Every side- A' parent square in $[0, L]^2$ is assigned, according to its orientation row, a concentric side- B'_r closed core. The source containment inequality remains strict after common scaling by λ , so each assigned core lies strictly inside its parent.

Universal coverage. The two complete exact replays establish that every assigned core has charge at least

$$\Gamma = 1,000,047,559 \cdot 10^{-9}.$$

Contradiction for eleven parents. If eleven side- A' parents with disjoint interiors fit in $[0, L]^2$, their strict interior closed cores are pairwise disjoint. Their total charge is therefore at least 11Γ but at most M , contradicting

$$11\Gamma - M = 428,125 \cdot 10^{-9} > 0.$$

Rescaling. A packing of eleven unit squares in side L/A' would scale by A' to eleven side- A' parents in side L . Hence no such packing exists. Compactness of the feasible packing space gives the strict inequality $s(11) > L/A'$.

7 Sanity checks and failure controls

The improvement is small enough that precision failure would be a natural concern. The computations therefore avoid floating geometry in the proof path. The factor λ is an exact rational number, and all new denominators are retained in exact rational/integer arithmetic. A deliberately larger shrink, $\lambda = 1 - 2 \times 10^{-7}$, fails quickly: rows 543–574 and 615–626 open substantially lower charge cells. Thus the verifier is demonstrably sensitive to small geometric

changes rather than rounding them away. In contrast, $\lambda = 1 - 10^{-9}$ survives the complete catalogue in both exact replays.

The source certificate itself is also reproduced correctly by the independent audit on known nonuniform rows, including the original global bottleneck row 11962 with charge 999,962,528. This is a useful guard against an implementation that would spuriously return a constant plateau for every row.

8 Public record, attribution, and scope

A public check on 29 September 2026 found that the Squares Project still listed Kleddamag’s $s(11) > 31/8$ as the strongest verified lower bound for eleven squares. No registered strict lower bound at the value proved here, or at a larger value, was located in that check. This statement records the result of the dated public search; the theorem itself depends only on the certificate and exact verification described above.

The present work is a derivative continuation of Kleddamag’s certificate and the Squares Project threshold-certificate framework. Those sources must be credited prominently. No claim is made to the source certificate, to Trump’s upper-bound construction, or to the general threshold-counting theorem. The new contribution is the explicit reweighting, the certified rational parent/core shrink, and the complete exact replay at the improved ratio.

9 Limitations and next steps

The bound is only

$$\frac{31}{7,999,999,992} \approx 3.875 \times 10^{-9}$$

above $31/8$. It should therefore be viewed as a proof-of-slack result: it shows rigorously that $31/8$ is not the endpoint of this frozen certificate architecture after reweighting. The next mathematical question is not whether the method crosses 3.875—it does—but how far the common shrink can be pushed before the first exact geometric cliff, and whether further reweighting can repair the newly exposed rows. None of that is needed for the theorem stated here.

10 Reproducibility package

The accompanying archive contains the frozen source and derived certificates, the recorded receipts, a portable Kleddamag-derived primary Python sweep, and an independent exact implementation. The release-level workflow is: run the static integrity audit; replay all 12,028 rows with the primary checker; replay them with the independent implementation; and compare the one-bin histograms and surplus. A quick smoke mode checks both implementations on row 0 before a full run. No numerical optimizer is needed to verify the final fixed certificate.

References and source record

1. Kleddamag. *Eleven unit squares: a certified lower bound*. Public GitHub repository, September 2026. Reproducible certificate for $s(11) > 31/8$. <https://github.com/Kleddamag/11-squares-certified-bound>

2. Joshua Levy and collaborators. *The Squares Project*. Public repository and $n = 11$ frontier record, accessed 29 September 2026. <https://github.com/jlevy/squares>
3. Walter Trump. 1979 eleven-square packing, as reconstructed and exactly verified in the Squares Project; it supplies the current best-known upper bound near 3.877083590022814.

A Exact headline data

$$\begin{aligned}
L &= 191/50, \\
A' &= 190999999809/193750000000, \\
\lambda &= 999999999/1000000000, \\
L/A' &= 3875000000/999999999, \\
\Gamma &= 1000047559 \times 10^{-9}, \\
M &= 11000095024 \times 10^{-9}, \\
11\Gamma - M &= 428125 \times 10^{-9}.
\end{aligned}$$

Rows: 12,028 (0–12,027).

Improved certificate SHA-256: 31e10ceb8368cc858e61f45ce7cfee783e8d0c7910893559164810f45439cc34.